

MUONIO LAUKUKERO, Finland

WMO: 028200

Lat: 68.063N

Lon: 24.033E

Elev: 762

StdP: 92.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.3	-24.3	0.5	-21.4	-21.6	0.6	-18.7	25.6	-11.0	23.6	-9.5	12.4	230	1.175

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	20.3	13.5	18.2	12.7	16.3	11.7	14.9	18.6	13.6	16.8	12.5	15.0	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	10.6	15.3	12.2	9.7	14.0	11.2	9.1	13.2	44.0	18.9	40.5	16.8	37.4	15.0	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.9	18.0	DB	-24.6	21.3	2.1	2.1	-26.1	22.8	-27.4	24.0	-28.6	25.2	-30.2	26.7	(4)
(5)			WB	-22.8	16.1	3.9	1.9	-25.6	17.4	-27.9	18.6	-30.1	19.6	-32.9	21.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-11.1	-10.8	-8.7	-4.3	1.9	6.8	11.1	9.1	4.6	-2.3	-6.0	-8.6	(6)	
	DBStd	8.60	4.23	4.72	3.59	3.62	5.16	4.50	4.23	3.45	3.28	3.42	3.32	3.67	(7)	
	HDD10.0	4322	653	583	581	428	257	118	38	59	162	383	481	578	(8)	
	HDD18.3	7237	912	817	839	678	509	347	228	288	411	641	731	837	(9)	
	CDD10.0	129	0	0	0	0	7	21	71	30	1	0	0	0	(10)	
	CDD18.3	3	0	0	0	0	0	0	3	1	0	0	0	0	(11)	
	CDH23.3	6	0	0	0	0	0	0	5	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	9.3	10.6	10.4	10.7	9.0	8.5	8.6	7.6	7.6	8.8	9.3	9.8	10.7	(14)	
(15) Precipitation	PrecAvg	512	31	26	23	26	31	55	80	74	50	44	39	34	(15)	
	PrecMax	683	58	57	45	65	79	104	147	126	93	88	80	61	(16)	
	PrecMin	391	10	4	8	10	5	3	27	36	10	16	11	10	(17)	
	PrecStd	82	10	13	11	12	20	27	34	28	22	20	18	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	0.2	-0.3	8.5	19.4	21.2	23.9	19.9	12.9	7.1	3.9	0.0	(19)	
		MCWB	-1.1	-0.5	-1.4	4.2	10.4	13.3	15.6	13.6	9.8	5.7	2.7	-0.1	(20)	
	2%	DB	-2.8	-2.4	-1.5	4.8	15.5	18.0	21.4	17.9	11.5	4.6	1.3	-1.5	(21)	
		MCWB	-3.0	-2.6	-2.0	2.0	10.0	11.9	14.1	13.0	9.4	4.0	1.0	-1.7	(22)	
	5%	DB	-4.0	-3.4	-2.5	2.2	12.1	15.8	19.4	15.7	10.3	3.7	-0.5	-2.9	(23)	
		MCWB	-4.2	-3.7	-2.9	0.5	7.7	10.5	13.5	11.8	8.5	3.2	-0.7	-3.0	(24)	
	10%	DB	-5.4	-4.6	-3.6	0.2	9.2	13.7	17.4	13.7	9.0	2.5	-1.7	-3.9	(25)	
		MCWB	-5.6	-4.8	-4.0	-0.8	6.0	9.3	12.6	11.1	7.9	2.0	-1.8	-4.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-0.1	-0.9	4.5	11.2	14.8	16.9	15.9	11.6	5.2	2.7	0.0	(27)	
		MCDB	-1.0	0.3	-0.7	8.1	16.6	17.3	19.5	17.9	11.9	5.4	2.9	0.0	(28)	
	2%	WB	-3.0	-2.6	-1.9	2.5	9.9	12.9	15.5	13.7	9.9	4.3	0.8	-1.7	(29)	
		MCDB	-2.8	-2.5	-1.6	4.5	15.1	15.1	19.3	16.1	11.0	4.6	1.6	-1.5	(30)	
	5%	WB	-4.2	-3.7	-2.9	0.6	8.3	11.1	14.4	12.6	8.9	3.3	-0.7	-3.0	(31)	
		MCDB	-4.1	-3.5	-2.6	1.9	12.0	13.7	18.3	14.7	9.8	3.7	-0.6	-2.9	(32)	
	10%	WB	-5.6	-4.8	-3.9	-0.6	6.0	9.6	13.4	11.6	8.0	2.0	-1.9	-4.0	(33)	
		MCDB	-5.4	-4.6	-3.6	0.2	9.0	12.9	16.7	13.5	8.7	2.4	-1.8	-3.9	(34)	
(35) Mean Daily Temperature Range	MDBR		3.5	3.3	3.4	4.0	5.5	6.3	6.1	5.2	4.0	3.1	3.2	3.2	(35)	
	5% DB	MCDBR	5.1	4.1	4.1	5.7	8.3	8.7	7.8	7.0	5.4	4.3	4.5	3.8	(36)	
		MCWBR	5.4	4.1	3.9	4.1	4.8	4.7	4.2	4.1	3.9	3.6	4.2	3.9	(37)	
	5% WB	MCDBR	5.0	4.0	3.9	5.1	8.0	7.2	7.1	5.9	4.5	3.8	4.4	3.8	(38)	
		MCWBR	5.2	4.0	3.8	3.9	5.2	4.6	4.4	3.8	3.7	3.4	4.2	4.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.168	0.226	0.262	0.289	0.307	0.312	0.319	0.306	0.279	0.245	N/A	N/A	(40)	
	taud		1.917	2.116	2.233	2.262	2.461	2.534	2.518	2.537	2.583	2.343	N/A	N/A	(41)	
	Ebn at Noon		128	593	788	864	891	895	874	850	793	583	N/A	0	(42)	
	Edh at Noon		8	45	74	97	90	86	85	75	55	36	N/A	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.04	0.49	1.99	3.94	4.91	4.97	4.65	3.35	1.78	0.59	0.09	0.00	(44)	
	RadStd		0.00	0.06	0.12	0.20	0.48	0.36	0.46	0.25	0.20	0.10	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page